

Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)			use of IR spectroscopy (1) peak at 650 to 800 (cm^{-1}) (1) OR use of mass spectrometry (1) peak at m/z 64/66 (allow 64.5) (1)		2				
	(b)			(mechanism involves) radical / homolytic fission (1) initiation $\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$ (1) propagation e.g. $\text{C}_2\text{H}_6 + \text{Cl}\cdot \rightarrow \cdot\text{CH}_2\text{CH}_3 + \text{HCl}$ (1) termination e.g. $\cdot\text{CH}_2\text{CH}_3 + \text{Cl}\cdot \rightarrow \text{CH}_2\text{ClCH}_3$ (1) for both propagation and termination mark to be awarded chloroethane must be a product allow appropriate second propagation step in place of termination	4			4		
	(c)			2.0 g ethane = 0.067 mol (1) produces 0.067 mol of chloroethane with mass of 4.3 g (1) percentage yield = $\frac{1.0}{4.3} \times 100 = 23$ (1) ecf possible		3		3	1 1	

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	(d)			two possible products e.g. $\text{CH}_2\text{ClCH}_2\text{Cl}$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ (1) with relevant explanation/equation e.g. two ethyl radicals combine / $\bullet\text{CH}_2\text{CH}_3 + \bullet\text{CH}_2\text{CH}_3 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ (1) allow appropriate propagation step award (1) for one product and one explanation		2		2		
				Question 8 total	4	7	0	11	2	0